The FORCE™ Platform Achieves Robust Knock Down of Toxic Human Nuclear *DMPK* RNA and Foci Reduction in DM1 cells and in Newly Developed hTfR1/DMSXL Mouse Model

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Joachim, *living with DM1*

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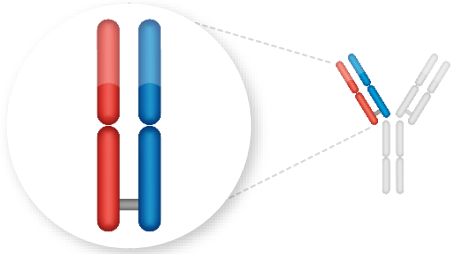
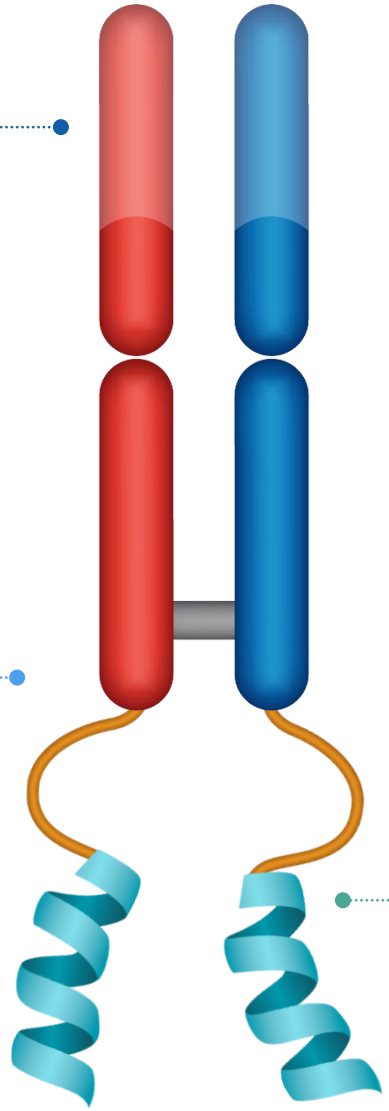
Dyne FORCE Platform: Modern Oligo Therapeutics for Muscle Diseases

ANTIBODY

Proprietary Fab targets TfR1 to enable muscle delivery



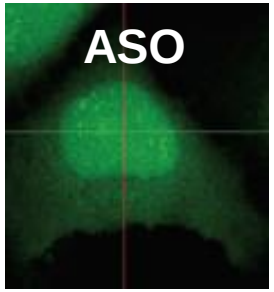


LINKER

Clinically validated, enables precise conjugation of multiple payloads to a single Fab

PAYLOAD

Modularity enables rational selection of payload to target the genetic basis of disease



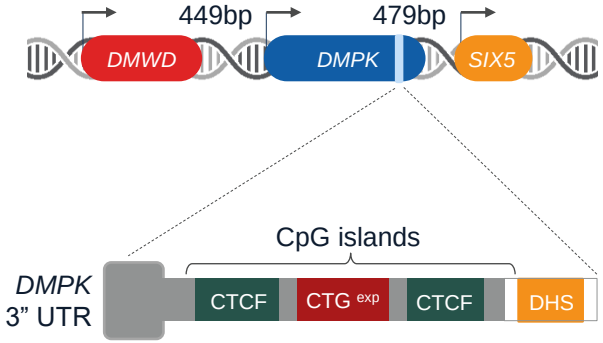
Nuclear localization

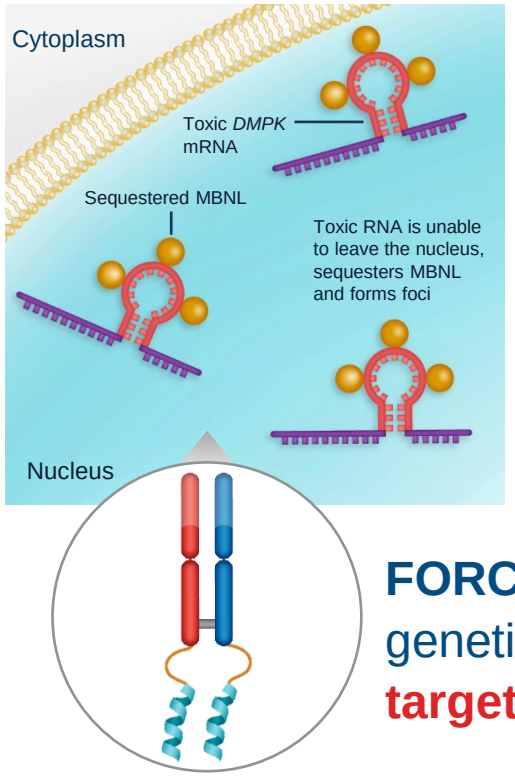
Cytoplasmic localization

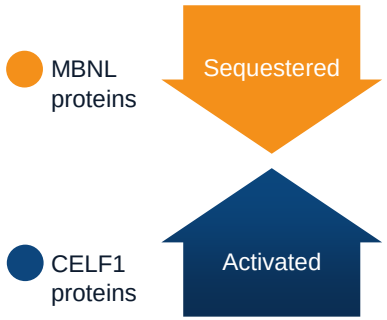
Adapted from: Ohrt et al: NAR , 2006, v.34, p.1369

FORCE Targets the Genetic Basis of DM1

DNA Triplet Repeats → **Toxic RNA Forms Foci** → **RNA Binds Splicing Proteins** → **Clinical Presentation**



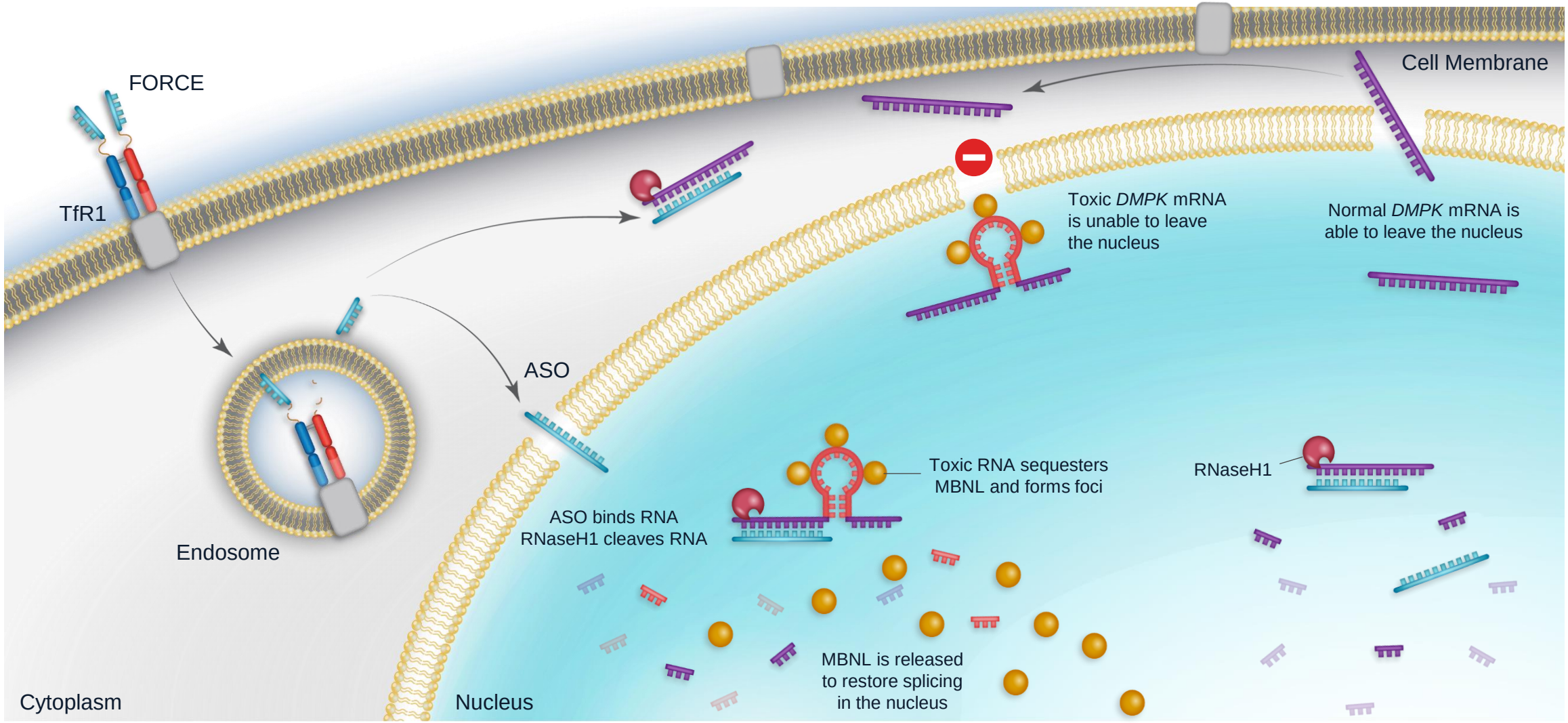




- Myotonia
- Muscle weakness
- Cardiac arrhythmia
- Pulmonary abnormalities

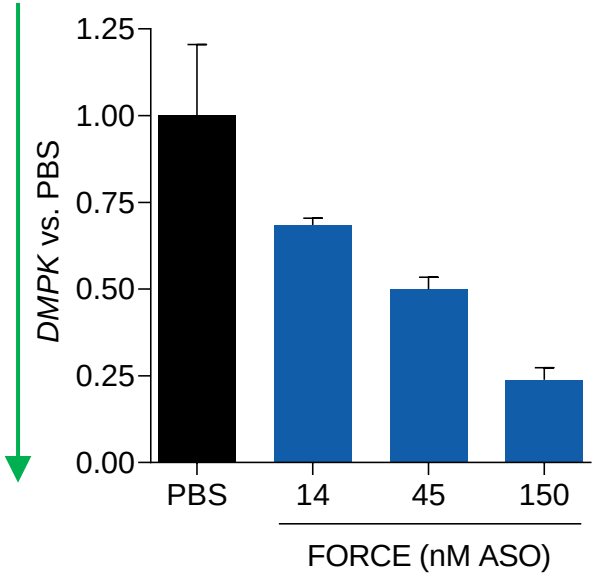
FORCE designed to address the genetic basis of disease by **targeting toxic nuclear *DMPK* RNA**

FORCE Targets Toxic Nuclear *DMPK* RNA

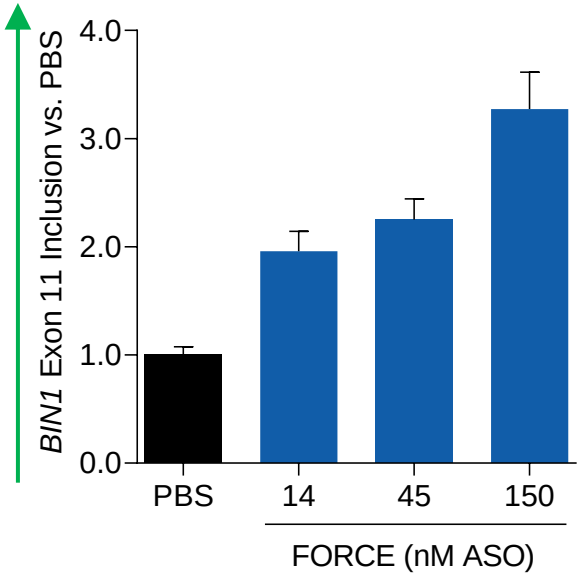


FORCE Conjugate Demonstrated Dose-dependent *DMPK* KD, Splicing Correction, and Foci Reduction in DM1 Myotubes with 380 CTG Repeats

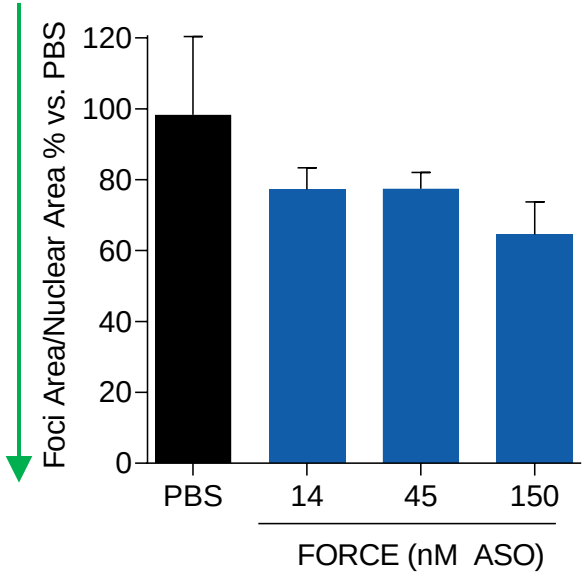
DMPK mRNA KD by qPCR



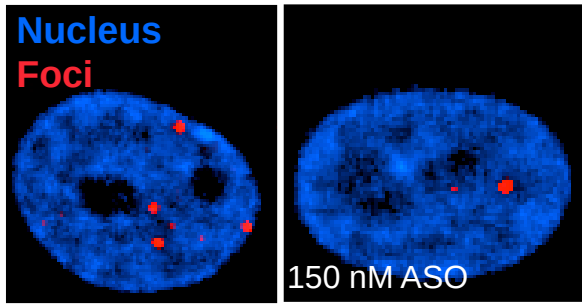
BIN1 splicing correction by qPCR



Toxic DMPK foci KD



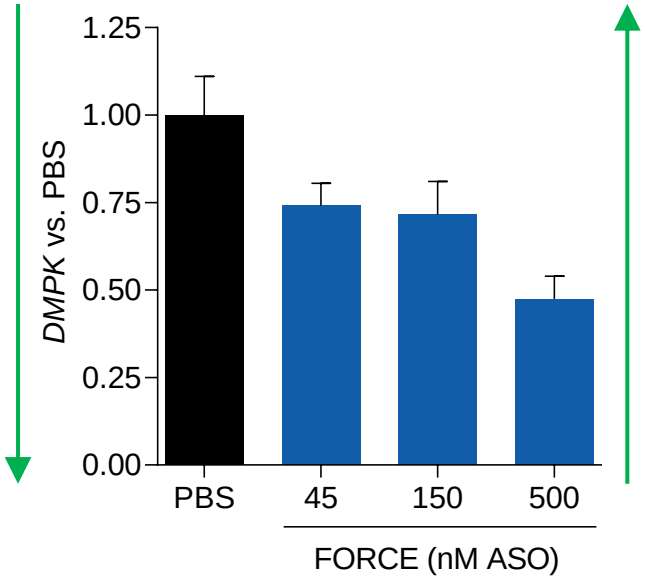
FISH with CAG PNA probe



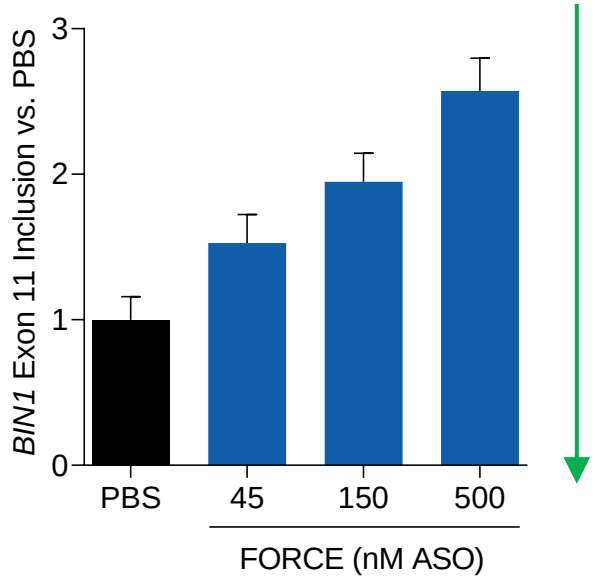
Note: Data are mean ±SD; n = 4

FORCE Conjugate Demonstrated Dose-dependent *DMPK* KD, Splicing Correction, and Foci Reduction in DM1 Myotubes with 2,600 CTG Repeats

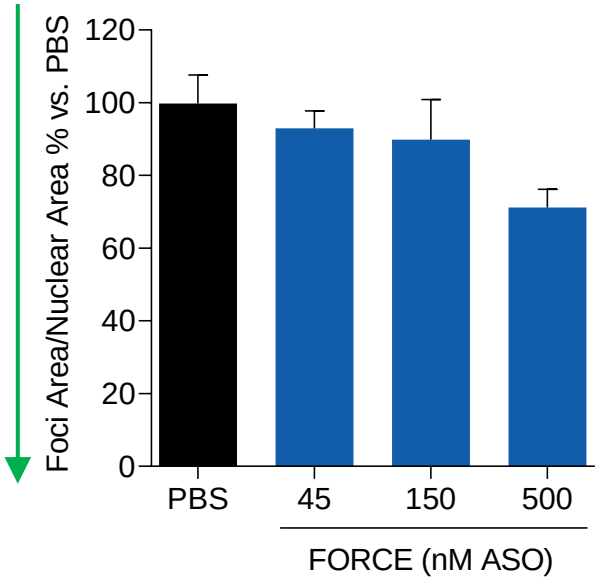
DMPK mRNA KD by qPCR



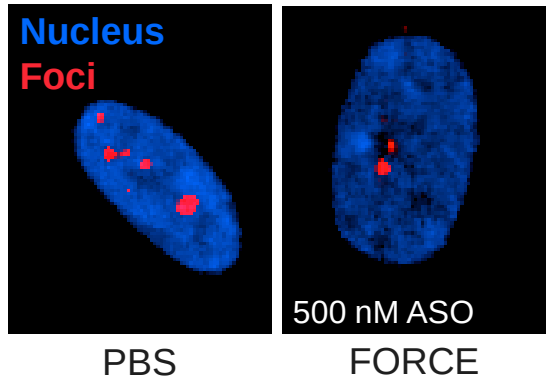
BIN1 splicing correction by qPCR



Toxic DMPK foci KD

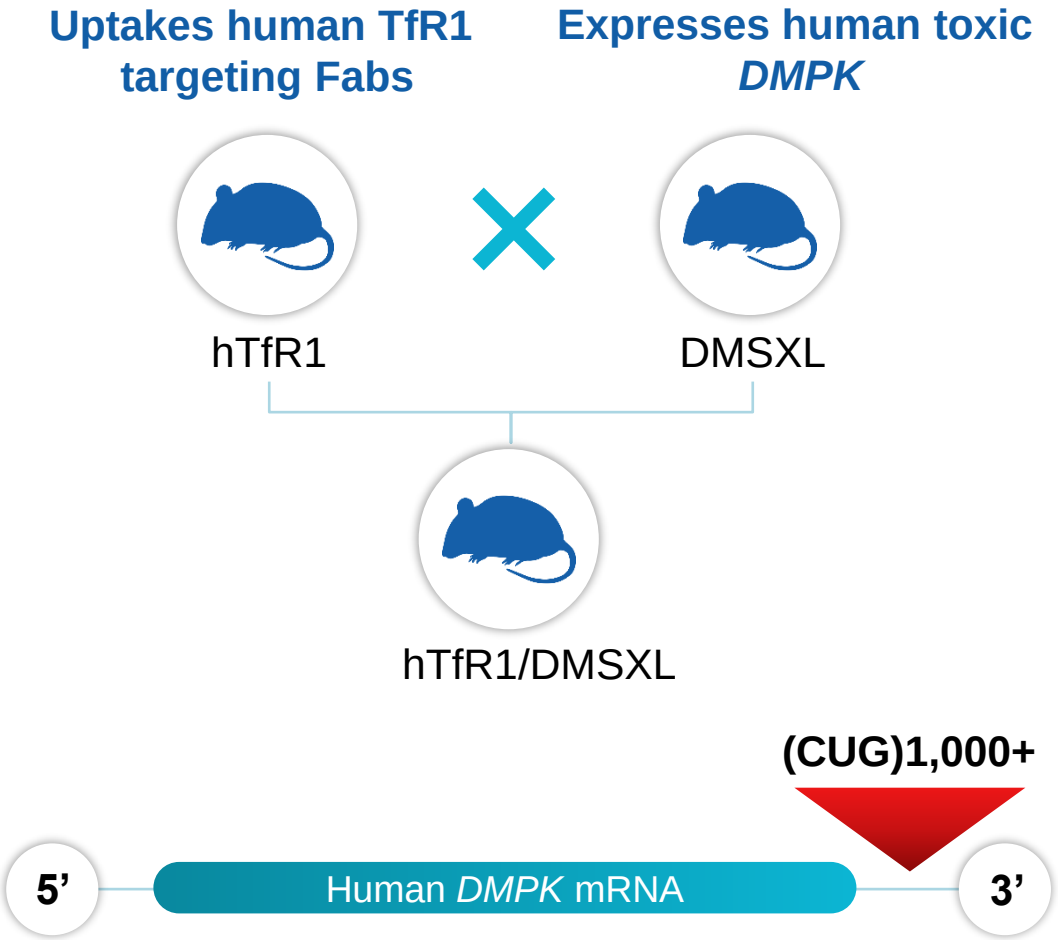


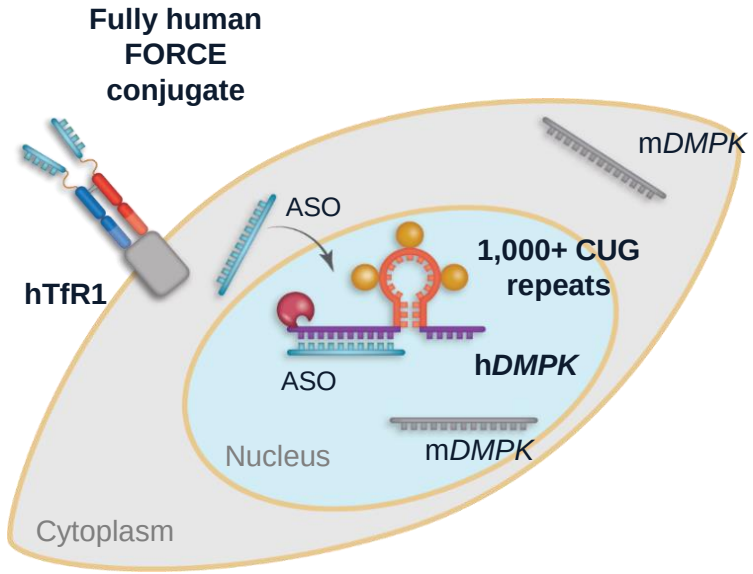
FISH with CAG PNA probe



Note: Data are mean ±SD; n = 4

hTfR1/DMSXL Mice: Innovative Model to Evaluate FORCE Conjugate Pharmacodynamics By Measuring Toxic Human Nuclear *DMPK* KD

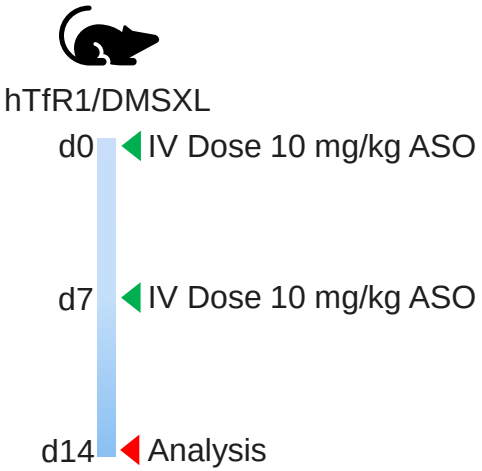


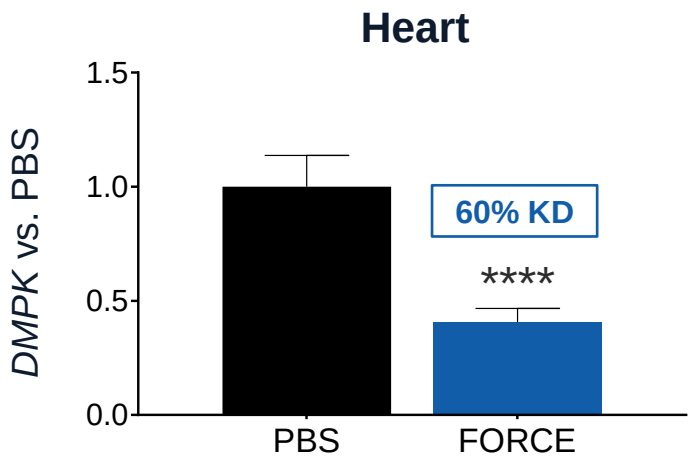


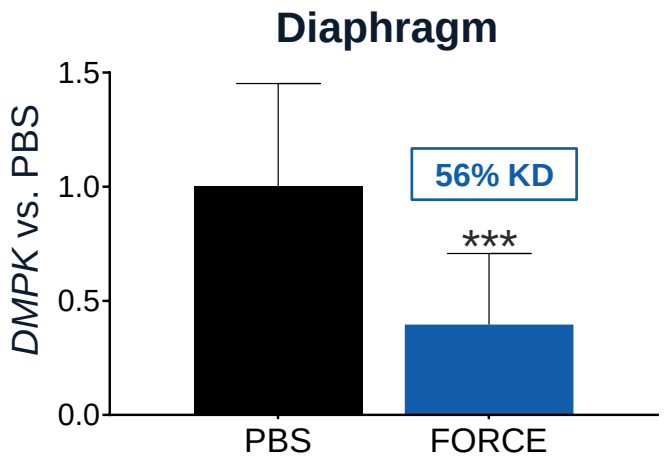
- Expresses human TfR1 receptor, enabling use of human TfR1-targeting Fabs
- Underestimates potency, expressing >10 times less human toxic *DMPK* vs. mouse *DMPK*

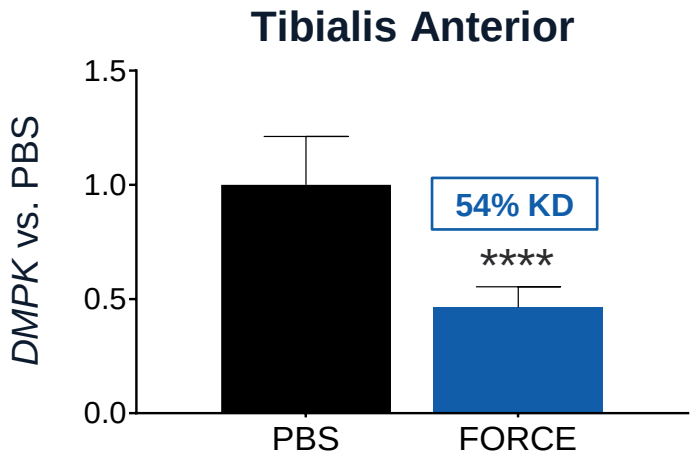
Note: DMSXL mice first described in PLOS Genetics 2012, 8(11):e1003043

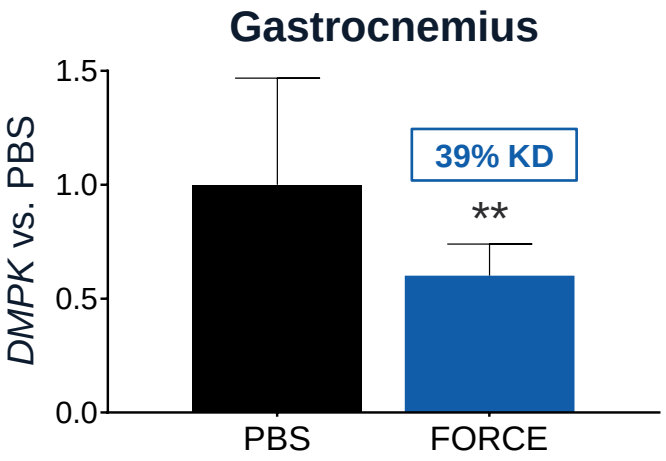
FORCE Conjugate Demonstrated Robust Toxic Human *DMPK* KD in hTfR1/DMSXL Mice after 14 Days







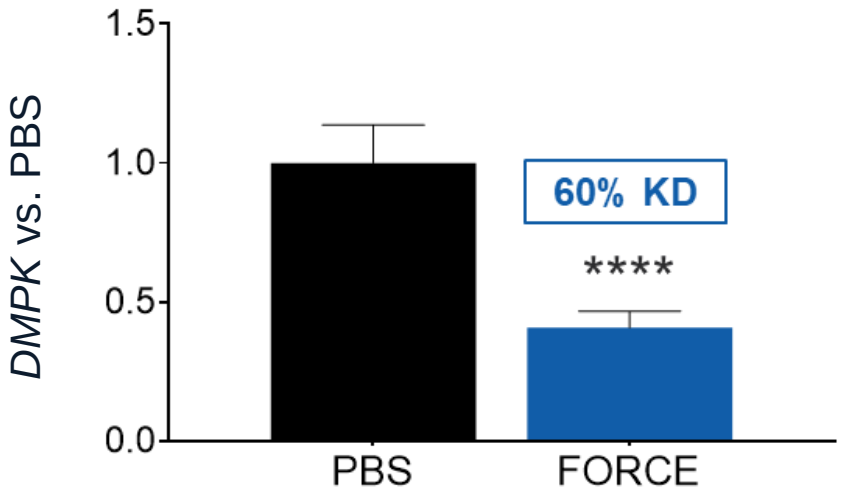




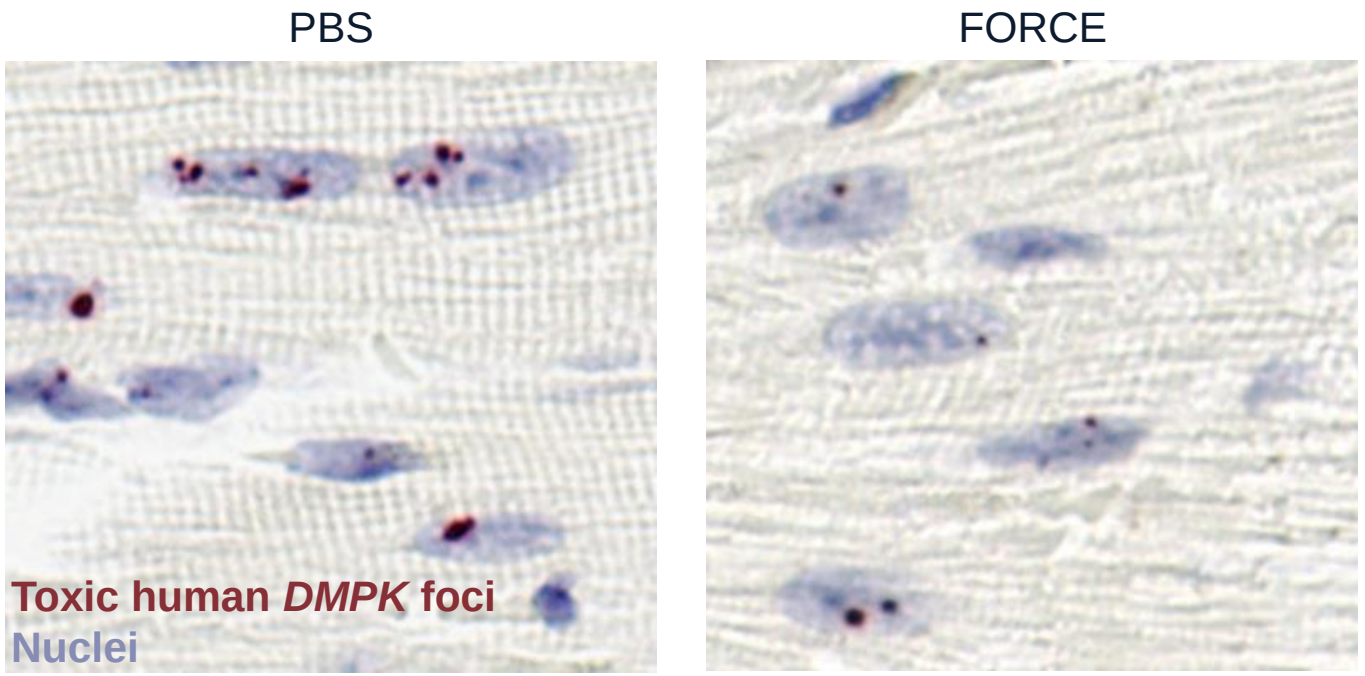
Note: Data are mean ±SD; n = 6 – 9; $^{**}p < 0.01$,$^{***}p < 0.001$, $^{****}p < 0.0001$, significant by *t*-test

FORCE Conjugate Demonstrated Robust Toxic Human *DMPK* mRNA and Foci KD in the Heart of hTfR1/DMSXL Mice

DMPK mRNA KD by qPCR



DMPK foci reduction in the heart by _in situ_ hybridization



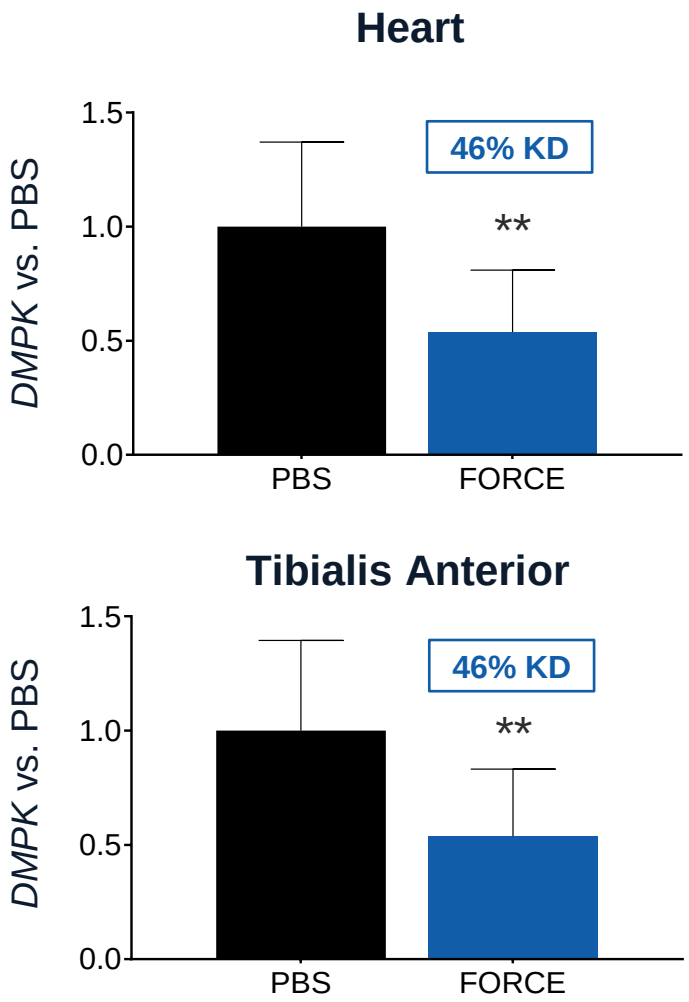
Note: 10 mg/kg ASO d0 and d7, analyzed d14; data are mean ±SD; n = 6 – 9; ****$p < 0.0001$, significant by *t*-test. Representative images of heart sections stained by RNAscope

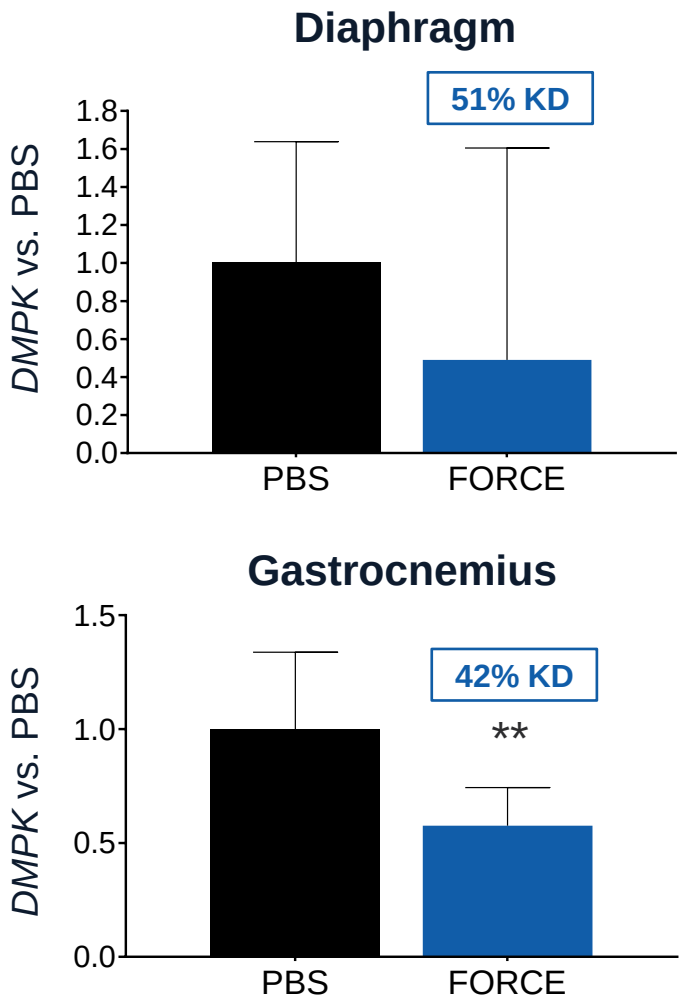
Single Dose of FORCE Conjugate Achieved Sustained Toxic Human *DMPK* KD at Week 4 in hTfR1/DMSXL Mice

hTfR1/DMSXL

Wk0 IV Dose 10 mg/kg ASO

Wk4 Analysis





Note: Data are mean ±SD; n = 6; ***p* < 0.01, significant by *t*-test

Conclusions

- FORCE is designed to address the genetic basis of DM1 by targeting toxic nuclear DMPK RNA
- These new data demonstrated that a fully human FORCE conjugate can:
  - Correct the DM1 phenotype of patient-derived myoblast cultures with a range of repeats, including those representative of severe DM1
  - Reduce toxic nuclear human *DMPK* foci in cardiac muscle of hTfR1/DMSXL mice
  - Lead to sustained KD of toxic nuclear human *DMPK* in hTfR1/DMSXL mice after a single dose
- These data strongly support further development of our DM1 therapeutic candidate, including a planned clinical study

Acknowledgements

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